Lecture 5

Tuesday, 24 January 2023 1:29 PM

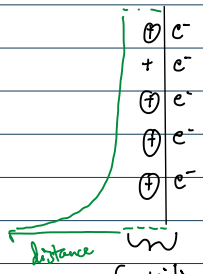
Double Layer

The region very near the electrode surface, where some charge separation occurs, but the charge built up will be exactly balanced by electrode

Due to brownian motion, this is not an exact single layer in solution, but instead net charge decreases exponentially away from electrode

↓

defined by debye length



Capacitive double layer

permittivity

$$\lambda = \sqrt{\frac{\varepsilon RT}{F^2 \sum_j z_j^2 c_{j,\infty}}}$$

; ε = (dielectric constant)(permittivity of free space)

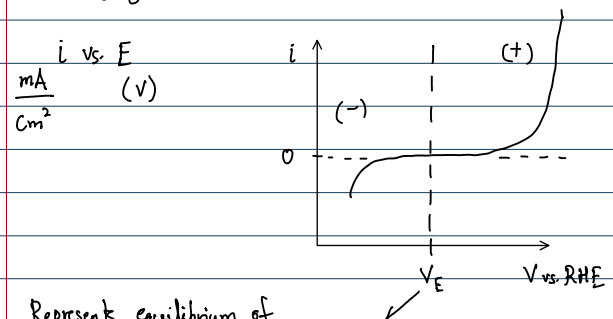
$K = 78$ (for water) $\varepsilon_0 = 8.854 \times 10^{-14}\ F/cm$

Electrochemical Rates

+ If we increase current density, we could reduce total # of devices needed

+ We need to understand the driving force required to achieve such an increase
  - ↳ deriving electrochemical rate laws

↳ Changing potential has an impact on the current → Varying potential is called polarizing an electrode



+ Voltage below & above V_{oc} leads to flow of charge one way vs other

+ Relationship b/w i & V appears exponential

+ $i \propto$ rxn rate (by Faraday's Law)

↓

Arrhenius equation ($k \propto e^{-E_a/RT}$) → originally an empirical relationship

Represents equilibrium of a single reversible electrode rxn

$[R \underset{k_c}{\overset{k_a}{\rightleftharpoons}} O + e^-]$

i.e. the rate of forward & reverse are equal resulting in 0 current

Same activation energy approach for echem rxns:

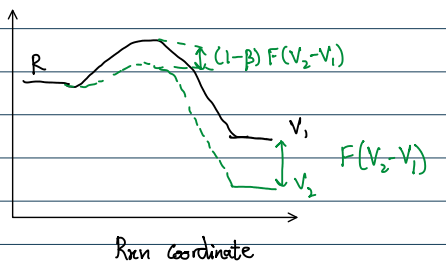
$W_{rev} = -F\Delta V$ ← work req in moving 1 mol e^- } ∝ Energy is associated

forward & reverse are equal resulting in 0 current @ the electrode

$r = k_a C_R - k_c C_o$

$W_{rev} = -F\Delta V$ ← work req in moving 1 mol e^- across energy barrier ΔV } ∼ Energy is associated w/ e^-

ex) $R \geq 0 + e^-$



β represents fraction of ΔV required to reach a new transition state $(0 \leq \beta \leq 1)$

So,

$$E_{A_{forward}}(V_2) = E_{A_{forward}}(V_1) - (1-\beta)F(V_2-V_1)$$

$$E_{A_{forward}}(V) = E_{A_{forward}}(0) - (1-\beta)FV \quad \text{(forward rxn is anodic)}$$

$$E_{A,a}(V) = E_{A,a}(0) - (1-\beta)FV \rightarrow \text{change in potential of 1 electrode (for can just replace w/ V)}$$

$$E_{A,c}(V) = E_{A,c}(0) + \beta FV \quad \text{(Cathodic)}$$

$$k_{forward} = A\exp\left[\frac{-E_{A,a}(0) + (1-\beta)FV}{RT}\right] \quad \text{(Arrhenius dependence)}$$

$$= k_a \exp\left[\frac{(1-\beta)FV}{RT}\right] \quad ; \quad k_a = \exp\left(\frac{-E_{A,a}}{RT}\right)$$

$$k_{rev} = k_c \exp\left[\frac{-\beta FV}{RT}\right]$$

$$\therefore \boxed{r = \frac{i}{F} = C_R k_a \exp\left[\frac{(1-\beta)FV}{RT}\right] - C_o k_c \exp\left[\frac{-\beta FV}{RT}\right]}$$